example of point mutation that results in the change of amino acid residue glutamic acid to valine (**Fig. 5.10**). The mutant haemoglobin undergoes polymerisation under oxygen tension causing the change in the shape of the RBC from biconcave to a sickle shaped structure.

The effect of point mutation can be understood by the following example.

ABC DEF GHI JKL

If we insert a letter **O** between **DEF** and **GHI** the arrangement would be

ABC DEF OGH IJK L

If we insert **OQ** at the same place the arrangement would be

ABC DEF OQG HIJ KL

The above information shows that insertion or deletion of one or two bases, changes the reading frame from the point of insertions or deletions. Such mutations are referred to as frame shift insertion or deletion mutations. This forms the genetic basis of proof that codon is a triplet and is read in a continuous manner

5.11. tRNA – the adapter molecule

The transfer RNA, (tRNA) molecule of a cell acts as a vehicle that picks up the amino acids scattered through the cytoplasm and also reads specific codes of mRNA molecules. Hence it is called an **adapter molecule.** This term was postulated by Francis Crick.

The two dimensional clover leaf model of tRNA was proposed by Robert Holley. The secondary structure of tRNA depicted in **Fig. 5.11** looks like a **clover leaf**. In actual structure, the tRNA is a compact molecule which looks like an **inverted L**. The clover leaf model of tRNA shows the presence of three arms namely DHU arm, middle arm and TΨC arm. These arms have loops such as amino acyl binding loop, anticodon loop and ribosomal binding loop at their ends. In addition it also shows a small lump called variable loop or extra arm. The amino acid is attached to one end (amino acid acceptor end) and the other end consists of three anticodon nucleotides. The anticodon pairs with a codon in mRNA ensuring that the correct amino acid is incorporated into the growing polypeptide chain. Four different regions of double-stranded RNA are formed during the folding process. Modified bases are especially common in tRNA. Wobbling between anticodon and codon allows some tRNA molecules to read more than one codon.

The process of addition of amino acid to tRNA is known as **aminoacylation** or **charging** and the resultant product is called aminoacyl- tRNA (charged tRNA). Without aminoacylation tRNA is known as uncharged tRNA **(Fig. 5.12).** If two such tRNAs are brought together peptide bond formation is favoured energetically. Numbers of amino acids are joined by peptide bonds to form a polypeptide chain. This aminoacylation is

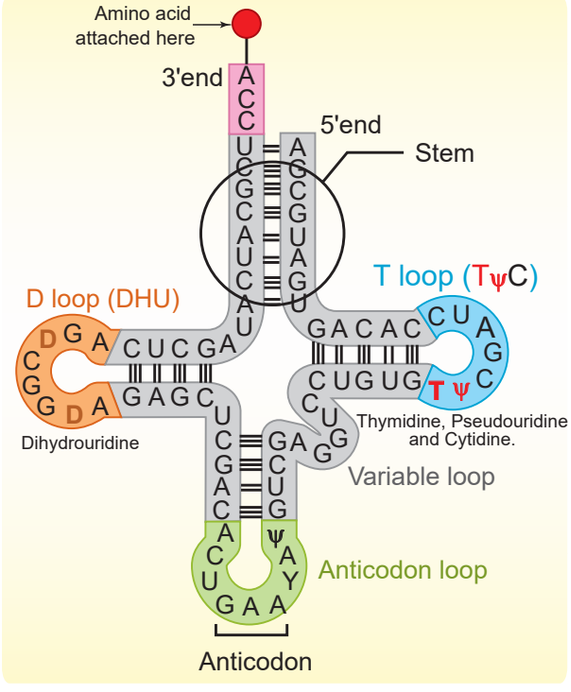


Fig. 5.11 Holley's two-dimensional clover leaf model of transfer RNA

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